

732A96/TDDE15 Advanced Machine Learning

Graphical Models

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Lecture 3: Parameter Learning

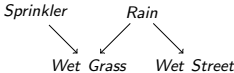
Contents

- ▶ Parameter Learning for BNs
 - ▶ Maximum Likelihood

Literature

- ▶ Main source
 - ▶ Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006. Chapter 8.
- ▶ Additional source
 - ▶ Koski, T. J. T. and Noble, J. M. A Review of Bayesian Networks and Structure Learning. *Mathematica Applicanda* 40, 51-103, 2012.

Parameter Learning for BNs: Maximum Likelihood

DAG	Parameter values for the conditional probability distributions
 <pre> graph TD Sprinkler --> WetGrass[Wet Grass] Rain --> WetGrass Rain --> WetStreet[Wet Street] </pre>	$q(s) = (0.3, 0.7) = (\theta_{s_0}, \theta_{s_1})$ $q(r) = (0.5, 0.5) = (\theta_{r_0}, \theta_{r_1})$ $q(wg r_0, s_0) = (0.1, 0.9) = (\theta_{wg_0 r_0, s_0}, \theta_{wg_1 r_0, s_0})$ $q(wg r_0, s_1) = (0.7, 0.3) = (\theta_{wg_0 r_0, s_1}, \theta_{wg_1 r_0, s_1})$ $q(wg r_1, s_0) = (0.8, 0.2) = (\theta_{wg_0 r_1, s_0}, \theta_{wg_1 r_1, s_0})$ $q(wg r_1, s_1) = (0.9, 0.1) = (\theta_{wg_0 r_1, s_1}, \theta_{wg_1 r_1, s_1})$ $q(ws r_0) = (0.1, 0.9) = (\theta_{ws_0 r_0}, \theta_{ws_1 r_0})$ $q(ws r_1) = (0.7, 0.3) = (\theta_{ws_0 r_1}, \theta_{ws_1 r_1})$ $p(s, r, wg, ws) = q(s)q(r)q(wg s, r)q(ws r)$

- ▶ In general,

$$q(X_i = k | Pa_i = j) = \theta_{X_i=k | Pa_i=j}$$

- ▶ Recall that

$$p(X_i = k | Pa_i = j) = q(X_i = k | Pa_i = j)$$

Parameter Learning for BNs: Maximum Likelihood

- Given a sample $d_{1:N}$, the log likelihood function is

$$\begin{aligned}\log p(d_{1:N}|\theta, G) &= \log \prod_l p(d_l|\theta, G) = \log \prod_l \prod_i p(d_l[X_i]|d_l[Pa_i], \theta) \\ &= \log \prod_l \prod_i \theta_{X_i=d_l[X_i]|Pa_i=d_l[Pa_i]} = \log \prod_i \prod_j \prod_k \theta_{X_i=k|Pa_i=j}^{N_{ijk}} \\ &= \sum_i \sum_j \sum_k N_{ijk} \log \theta_{X_i=k|Pa_i=j}\end{aligned}$$

where N_{ijk} is the number of instances in $d_{1:N}$ with $X_i = k$ and $Pa_i = j$.

- To maximize the log likelihood function subject to the constraint $\sum_k \theta_{X_i=k|Pa_i=j} = 1$ for all i and j , we maximize

$$\sum_i \sum_j \sum_k N_{ijk} \log \theta_{X_i=k|Pa_i=j} + \sum_i \sum_j \lambda_{ij} (\sum_k \theta_{X_i=k|Pa_i=j} - 1)$$

where λ_{ij} are called Lagrange multipliers.¹

- Setting to zero the derivative with respect to $\theta_{X_i=k|Pa_i=j}$ gives

$$\theta_{X_i=k|Pa_i=j} = -N_{ijk}/\lambda_{ij}$$

- Replacing in the constraint gives $\lambda_{ij} = -N_{ij}$ and $\theta_{X_i=k|Pa_i=j}^{ML} = N_{ijk}/N_{ij}$.

¹Any stationary point of the Lagrangian function is a stationary point of the original function subject to the constraints, which is a maximum of the original function subject to the constraints because the log likelihood function is concave.

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Thank you